



Hybrid-domain audio watermarking using Simplifying Graph Convolution

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ABSTRACT

As low-artifact deepfake audio proliferates, the credibility of audio in personal, legal, and public communications is severely undermined, creating an urgent demand for reliable audio watermarking solutions. Current mainstream audio watermarking methods are mostly non-graph-based and struggle to strike a good balance between imperceptibility and robustness against various attacks. Meanwhile, graph-based methods, which have shown advantages in capturing structural features and enhancing robustness in other fields (e.g., audio forgery detection and image watermarking), suffer from high computational overhead when applied to audio watermarking due to nonlinearities between consecutive layers. To address these limitations, we propose an improved hybrid-domain audio watermarking framework that uses Simplifying Graph Convolution (SGC) to optimize the original graph attention network, integrating Discrete Wavelet Transform (DWT), Graph-Based Transform (GBT), and Singular Value Decomposition (SVD) to enhance robustness while ensuring imperceptibility. The framework first decomposes the audio signal into multi-scale subbands via DWT, then extracts structural features using GBT and SVD, and finally refines features through SGC to reduce computational complexity while enhancing robustness. Specifically, we adopt linear SGC feature aggregation (in graph domain) to replace complex nonlinear graph transformations (reducing computational complexity while retaining the ability to capture critical spectro-temporal cues), embed watermarks in the low-frequency band of audio via DWT (in frequency domain), and utilize GBT-SVD (in matrix domain) to capture structural features for enhanced anti-attack performance. Extensive experimental results demonstrate that the proposed hybrid-domain watermarking method achieves a superior balance between imperceptibility and robustness compared to existing solutions, with higher computational efficiency and stable performance across different audio attributes and attack types.

1. Introduction

The rapid evolution of multimedia technology and artificial intelligence has made digital audio a dominant medium in modern information transmission. Nevertheless, the emergence and widespread abuse of deepfake audio have severely threatened the credibility of audio content. In the digital era, audio is indispensable in personal, legal, public communications and remote services (Nasr et al., 2025), yet audio forgery technologies and deep learning methods have undermined its credibility via low-artifact deepfake audio, demanding reliable anti-forgery solutions (Wang et al., 2017; Doan et al., 2023; Gao et al., 2024). As a typical proactive anti-forgery technology, audio watermarking has become a core research direction in the field of digital audio security. It embeds imperceptible and tamper-resistant information into audio signals to achieve authenticity verification and

tampering tracing, which is fundamentally different from passive audio forgery detection technologies (Uddin et al., 2024; Chauhan and Rizvi, 2013). Unlike detection systems that only identify forgeries, watermarking provides “positive authentication”: if the watermark is missing or altered, the audio integrity can be directly deemed compromised (Kosta et al., 2022). It should be clarified that current mainstream audio watermarking methods are non-graph-based. Early non-graph-based watermarking methods focused on time-domain or simple frequency-domain embedding, while modern deep learning-driven schemes have significantly improved robustness against common attacks (Chen et al., 2024).

In recent years, deep learning-based audio watermarking has become a research hotspot, with several representative schemes proposed:

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Robust-DNN (Kosta et al., 2022) pioneered a deep neural network-based embedding method in the Short-Time Fourier Transform domain, balancing robustness against removal attacks and audio quality but lacking stability under re-recording or noise attacks; WavMark (Chen et al., 2024) adopted Invertible Neural Networks for synchronous encoder-decoder training, improving embedding capacity while maintaining imperceptibility, yet its robustness degrades significantly under frequency-domain attacks; AudioSeal (Uddin et al., 2024) achieved strong resistance to adversarial and common attacks but compromised slight imperceptibility due to high embedding strength; DeAR (Liu et al., 2023) embedded watermarks in the low-frequency domain to resist re-recording distortion yet failed to leverage structural feature redundancy for enhanced robustness; DRAW (Li et al., 2024) used a dual-decoder structure to improve anti-desynchronization and replay performance but had higher computational overhead than single-decoder designs. Despite these advances, balancing imperceptibility and robustness remains a core challenge for non-graph-based audio watermarking (Liu et al., 2023; Li et al., 2024), driving continuous innovation in watermarking design. Timbre (Liu et al., 2024) introduces timbre watermarking to protect voice identity and intellectual property against voice cloning, achieving time-agnostic embedding and strong robustness to cloning-induced temporal distortions.

Against this backdrop, graph-based methods have attracted attention due to their unique advantages in capturing structural features and long-range dependencies. Such methods have been successfully applied in related fields. First, in audio forgery detection, graph-based schemes (Tak et al., 2021, 2022a,b) improve robustness to unseen forgeries by capturing complex spectro-temporal relationships. Moreover, in image watermarking, graph-based features are used to enhance anti-attack performance, such as Wen et al.'s algorithm (Wen et al., 2022), which combines Graph-Based Transform (GBT) and Singular Value Decomposition (SVD) (Hou et al., 2016; Golub and Van Loan, 2013) to capture structural features and resist watermark attacks. However, directly applying graph-based methods to audio watermarking faces critical limitations: their complex graph construction and iterative nonlinear updates lead to high computational complexity and slow inference speed, which are incompatible with the real-time processing requirements of audio watermarking.

Additionally, cross-modal technology transfer, from image watermarking to audio watermarking, is hindered by signal-specific structural differences. For example, Tavakoli et al.'s CNN method (Tavakoli et al., 2023) uses Discrete Wavelet Transform (DWT) (Mallat, 1989) for frequency decomposition to resist compression and filtering attacks, but direct adaptation to audio is ineffective due to differences in signal structure.

Existing graph-based audio watermarking methods rely on multi-layer graph convolution networks, which introduce heavy computational costs and limit practical deployment. Different from these methods, this work introduces simplifying graph convolution (SGC) into audio watermarking for the first time. The proposed SGC-based refinement removes redundant nonlinear transformations while preserving essential structural information, significantly reducing model complexity. Moreover, our hybrid-domain framework is the first to integrate frequency decomposition, graph structure modeling, and matrix decomposition with SGC, effectively addressing the limitations of single-domain and traditional graph-based methods.

To address the limitations of existing non-graph-based audio watermarking (poor balance between performance and efficiency) and the bottlenecks of direct application of graph-based methods (high computational overhead), this paper proposes a synergistic hybrid-domain audio watermarking framework integrating SGC-optimized feature processing and DWT-GBT-SVD fused embedding. Its key technical innovations include three aspects:

First, we design a DWT-GBT-SVD fused watermarking method for the hybrid-domain framework: DWT decomposes audio into frequency

bands (frequency domain) to enable low-frequency watermark embedding (ensuring imperceptibility), GBT converts audio segments into graph structures (graph domain) to capture structural redundancy, and SVD modifies singular values (matrix domain) for robust embedding (enhancing anti-attack performance).

Second, we adopt Simplifying Graph Convolution (SGC) (Wu et al., 2019) to optimize graph-domain feature processing in the hybrid-domain framework, replacing cascaded nonlinear graph transformation layers with linear feature mapping, which retains the ability to capture structural features while reducing computational complexity.

Third, we integrate SGC into the hybrid-domain watermarking module, replacing complex GBT graph construction with linear SGC aggregation to achieve efficient cross-domain feature extraction (from frequency domain to graph domain), reducing overhead while ensuring the structural feature stability required for SVD-based embedding. In terms of computational efficiency, the proposed SGC-based framework significantly reduces inference overhead. Experimental results show that compared with traditional graph convolution-based methods. The overall complexity is comparable to lightweight frequency-domain methods while maintaining stronger robustness. These results demonstrate that the proposed method achieves a favorable trade-off between performance and computational cost.

In this work, discrete wavelet transform (DWT) is adopted for audio signal decomposition. Compared with other wavelet transforms such as continuous wavelet transform (CWT) and stationary wavelet transform (SWT), DWT offers optimal time-frequency localization capability for audio signals, which can effectively separate the low-frequency core information and high-frequency noise components of audio. Meanwhile, DWT has relatively low computational overhead and good compatibility with subsequent graph-based transform (GBT) and singular value decomposition (SVD) operations. This characteristic enables the proposed framework to achieve a better balance between imperceptibility, robustness, and computational efficiency, making it more suitable for practical audio watermarking applications.

Different from existing single-domain audio watermarking methods and simple multi-technique combination schemes, this work presents two core scientific novelties. First, a novel hybrid-domain watermarking framework is constructed, which organically integrates frequency-domain decomposition, graph-based structural feature extraction, and matrix-domain embedding. This multi-dimensional integration breaks through the performance bottleneck of single-domain methods and jointly optimizes imperceptibility and robustness. Second, simplifying graph convolution (SGC) is introduced into the audio watermarking field for the first time, replacing traditional complex graph convolution operations. This design effectively reduces redundant graph calculations, lowers the overall computational complexity of the framework, and enhances the adaptability of the model to complex audio signals, which provides a new technical idea for efficient audio watermarking.

The robustness improvement of the proposed framework stems from the multi-dimensional constraint mechanism formed by the hybrid-domain integration. DWT filters high-frequency noise interference, GBT and SVD extract structural features with strong anti-distortion ability, and SGC further enhances the stability of structural features by simplifying graph operations. This multi-layered protection mechanism makes the watermark resistant to various common attacks.

Our main contributions are summarized as follows:

- We design a DWT-GBT-SVD fused audio watermarking method that synergistically leverages low-frequency imperceptibility (DWT), structural feature redundancy (GBT), and singular value stability (SVD). It achieves good imperceptibility and extraction accuracy and in balanced performance with robustness.
- We design an SGC-based cross-domain refinement method for audio watermarking, replacing the complex GBT with linear Simplifying Graph Convolution to realize efficient frequency-domain feature aggregation, reducing computational overhead while enhancing the stability of SVD-based watermark embedding.

- Extensive experiments on multiple datasets demonstrate the effectiveness of the proposed framework in accuracy, efficiency, and robustness across diverse attack types and audio attributes.

The remaining of this paper is organized as follows: Section 2 reviews related work in detail, contextualizing our optimizations with existing detectors and methods. Section 3 details the DWT-GBT-SVD fused watermarking method, with emphasis on SGC-based feature aggregation. Section 4 conducts comprehensive experiments to watermarking performance (imperceptibility/robustness) and computational efficiency. Section 5 concludes with key findings, limitations, and future directions.

2. Related work and preliminaries

In this section, we comb through core related models that lay the foundation for our research to clarify the technical background and innovative positioning of the proposed model. These models mainly cover four categories, including Simplifying Graph Convolution (SGC), audio watermarking technology, and image watermarking technology. The following subsections will elaborate on their structural features, core advantages, and respective roles in our research.

2.1. Simplifying Graph Convolution (SGC)

SGC is a lightweight Graph Convolutional Network (GCN) variant proposed to solve the problems of over-parameterization and high computational complexity of traditional GCNs (Wu et al., 2019). It simplifies multi-layer graph convolution into a single-layer linear transformation followed by activation, removing inter-layer non-linearities and collapsing multi-layer operations into a unified linear projection. The key formula of SGC is

$$X_{\text{out}} = S^K X_{\text{in}} \Theta \quad (1)$$

where S is the normalized adjacency matrix, K is the receptive field size (controlling K -hop neighbor aggregation), X_{in} and X_{out} are the input and output feature matrices, respectively, and Θ is the merged weight matrix.

2.2. Audio watermarking technology

2.2.1. WavMark model

WavMark is an end-to-end audio watermarking model designed for robust and imperceptible watermark embedding (Chen et al., 2024). Unlike traditional methods relying on manual feature design, it leverages a deep learning architecture based on CNNs and transformers to learn adaptive embedding strategies from large-scale audio datasets. Its core workflow and key technologies are as follows:

(i) STFT-Based Spectrogram Conversion: Time-domain audio $x_{\text{wave}} \in \mathbb{R}^L$ is converted to a spectrogram X_{spec} via Short-Time Fourier Transform (STFT), which separates the signal into magnitude and phase components. STFT applies the Fast Fourier Transform (FFT) to overlapping windows, mathematically expressed as

$$X_{\text{spec}}(\tau, \omega) = \int x(t)w(t - \tau)e^{-j\omega t} dt \quad (2)$$

where $w(t)$ is the window function, τ is the time offset, and ω is the angular frequency. The magnitude component is used for watermark embedding due to the phase's high sensitivity to human perception.

(ii) Watermark Embedding and Reconstruction: A binary watermark vector $\mathbf{m}_{\text{vec}} \in \{0, 1\}^{128}$ is expanded to a feature map $\mathbf{M}_{\text{spec}}$ matching X_{spec} 's magnitude component, then embedded via additive modification

$$X'_{\text{spec}} = X_{\text{spec}} + \gamma \cdot \mathbf{M}_{\text{spec}} \quad (3)$$

where γ is the embedding strength balancing robustness and imperceptibility. Inverse STFT (ISTFT) is applied to X'_{spec} to generate watermarked audio x'_{wave} .

(iii) Watermark Extraction with Invertible Neural Networks (INNs): For attacked audio, the extraction process reverses the embedding steps using an INN to recover $\hat{\mathbf{m}}_{\text{vec}}$. INNs feature an invertible forward pass (input and output dimensions are identical), enabling perfect input reconstruction (up to numerical precision)—a critical property for reversible watermark extraction. An INN consists of stacked invertible modules with parameter-sharing between encoding and decoding phases. For the l th module, the encoding process is

$$\begin{cases} \mathbf{a}_l = f_{\theta}(\mathbf{x}_l) \\ \mathbf{x}_{l+1} = \mathbf{x}_l \odot \sigma(\mathbf{a}_l) + \mathbf{m}_l \odot (1 - \sigma(\mathbf{a}_l)) \\ \mathbf{m}_{l+1} = \mathbf{m}_l \odot \sigma(\mathbf{a}_l) + \mathbf{x}_l \odot (1 - \sigma(\mathbf{a}_l)) \end{cases} \quad (4)$$

where $\sigma(\cdot)$ is the sigmoid function, $\odot$ denotes element-wise multiplication, and $f_{\theta}(\cdot)$ is a learnable function; the decoding process reverses these operations.

(iv) Loss Function Design: To balance multi-objectives, WavMark adopts three core loss functions and a weighted total loss. First, Audio Quality Loss constrains perceptual similarity between original and watermarked audio using L2 loss. Then we define the Message Recovery Loss, which ensures accurate watermark extraction using L2 loss between original and decoded watermarks. Next, Adversarial Loss, which enhances imperceptibility by deceiving a discriminator, is defined by. Finally, we define the Total Loss as a weighted combination of the three aforementioned.

As a baseline for watermarking experiments, WavMark's original performance provides a reference for verifying the effectiveness of improved models such as WavMark-DWT, WavMark-GBT, and WavMark-DGS.

2.2.2. AudioSeal model

AudioSeal is a real-time-oriented robust audio watermark model widely used in scenarios requiring dynamic watermark length (Chauhan and Rizvi, 2013). Its core advantages include: (i) integration of a Variable-Length Watermark Decoder (VLPD), supporting dynamic adjustment of watermark length without retraining the model; (ii) low computational complexity, with embedding and extraction latency less than 10 ms, suitable for low-power devices; (iii) basic robustness against common attacks such as resampling and MP3 compression.

However, its default watermark embedding position is incompatible with GBT-SVD embedding logic, limiting the optimization effect of GBT-SVD-based methods. In subsequent experiments, AudioSeal serves as a baseline to compare the performance of improved models such as AudioSeal-DWT, AudioSeal-GBT, and AudioSeal-DGS.

2.3. Timbre model

Timbre is a timbre-oriented audio watermarking method proposed in 2024 to address voice identity theft and timbre right infringement in generative AI era. Traditional detection-based anti-cloning methods suffer from distribution shift, while conventional watermarking struggles with temporal irregularities in cloning. To tackle these issues, Timbre embeds watermarks into timbre characteristics via time-agnostic embedding based on STFT repetition and temporal averaging, and integrates a cloning-aware noise layer for robust training. Extensive experiments show that it maintains high imperceptibility, achieves 100% extraction under 90% cropping, and delivers near-perfect recovery across diverse cloning pipelines, languages, and low-quality synthesis scenarios.

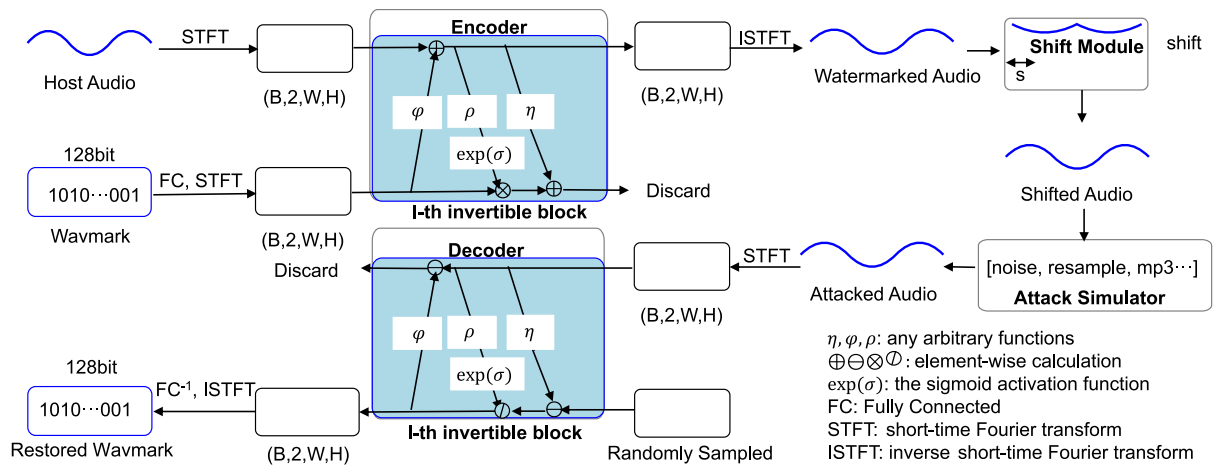


Fig. 1. Proposed audio watermarking detection method using DWT. The framework consists of watermark embedding, transmission, and detection stages. The discrete wavelet transform (DWT) decomposes the input audio into four subbands, and the watermark is embedded into the robust LL subband. After possible signal attacks during transmission, the hidden watermark is extracted from the received audio for authentication.

2.4. Image watermarking technology

2.4.1. Discrete Wavelet Transform (DWT)-based method

The DWT-based image watermarking method is a classic frequency-domain technique (Tavakoli et al., 2023). First, its core workflow relies on three key steps: the first step is DWT, which decomposes the image into Low-Low (LL, approximate information), Low-High (LH, horizontal detail), High-Low (HL, vertical detail), and High-High (HH, diagonal detail) sub-bands; the second step is watermark embedding, which embeds the watermark into the original LL sub-band coefficient; the third step is Inverse DWT (IDWT) reconstruction. The key insight for this study is that DWT's frequency separation capability can be extended to audio signals, which further provides a basis for designing audio watermark embedding positions.

2.4.2. Graph-Based Transform (GBT) and Singular Value Decomposition (SVD) method

The GBT-SVD-based image watermarking method combines the feature learning capability of Graph-Based Transform (GBT) and the matrix stability of Singular Value Decomposition (SVD) (Wen et al., 2022). First, its core workflow includes three steps. The first step is GBT feature extraction, which converts the image into a graph and extracts high-dimensional structural features via graph transformation. The second step is SVD embedding, which realizes watermark embedding through SVD. The third step is watermark extraction, which re-computes SVD on the watermarked feature matrix and recovers the watermark by comparing the diagonal matrix of singular values and the modified matrix. This method's capability to capture structural features further provides inspiration for enhancing the robustness of audio watermarks.

3. Audio watermark detection

3.1. Overall architecture

The framework for audio watermark detection has a structure illustrated in Fig. 1, and it mainly comprises three core components, a reversible encoder/decoder, a shift module, and an attack simulator. These components adopt an end-to-end training method, which enables seamless integration between them and joint optimization of the entire framework to enhance its overall performance. The core design concept of the framework is to embed watermarks into the frequency domain of audio that exhibits both robustness (to resist various audio processing attacks) and imperceptibility (to avoid affecting human perception of

audio quality), and this embedding is achieved using methods such as Discrete Wavelet Transform (DWT) or Graph-Based Transform-Singular Value Decomposition (GBT-SVD).

To optimize cross-domain feature processing efficiency and enhance watermark robustness, we further introduce an SGC-based cross-domain refinement method that replaces Graph-Based Transform (GBT) by Simplifying Graph Convolution (SGC) for linear and efficient aggregation of frequency-domain structural features. The framework then relies on the reversibility of Invertible Neural Networks (INN) to ensure that the decoder can accurately extract the embedded watermarks from audio that has undergone attacks. This forms a closed processing loop of “audio decomposition → watermark embedding → reversible encoding → attack resistance → reversible decoding → watermark recovery”, which ensures the reliability and effectiveness of the audio watermark detection process.

3.2. Reversible encoder/decoder

The reversible encoder/decoder is the core component of the audio watermark detection framework, and its main responsibilities include completing the embedding of watermark information into the original audio and extracting the watermarks from audio that has been subjected to attacks. The workflow of this encoder/decoder is closely integrated with four frequency-domain embedding methods, namely the Short-Time Fourier Transform (STFT) method used in WavMark, DWT, GBT-SVD, and the combined DWT-GBT-SVD method. Throughout the workflow, the input and output dimensions of each module are kept consistent to ensure the reversibility of the encoder/decoder, which is essential for accurate watermark extraction.

3.2.1. Audio representation

The network of the framework takes mono audio with a sampling rate of 16 kHz as the main input, and the Encoding Unit Length (EUL) is set to 1 s. As a result, the original input audio is represented as a 1-dimensional waveform vector with a length of 16,000, denoted as $x_{\text{wave}} \in L$ where $L = 16000$ is the number of sampling points in the audio. To enable watermark embedding in the frequency domain, the 1-dimensional waveform vector is further converted into a spectrogram or a sub-band matrix using techniques such as STFT or DWT, and the watermark is then embedded in the corresponding frequency-domain representation. The four frequency-domain embedding methods are detailed as follows.

Method 1:

The STFT-Based Embedding Method follows the design of the original WavMark model and uses STFT to convert the time-domain audio waveform into the frequency domain for watermark embedding. The time-frequency conversion stage adopts the core STFT logic in Section 2 to convert time-domain audio to spectrogram (refer to Eq. (2)). The STFT process is mathematically expressed as

$$X_{\text{spec}} = \text{STFT}(x_{\text{wave}}) \quad (5)$$

where $X_{\text{spec}} \in \mathbb{R}^{B \times C \times W \times H}$ is the frequency-domain feature map obtained from the STFT, B denotes the batch size, $C = 2$ (representing the magnitude and phase values of the frequency-domain signal respectively), W denotes the time dimension, and H denotes the frequency dimension.

In this method, watermarks are embedded into the magnitude component of X_{spec} because the phase component is more sensitive to human perception—modifying the phase component would more easily lead to noticeable changes in audio quality. The embedding formula for the watermark is

$$X'_{\text{spec}} = X_{\text{spec}} + \gamma \cdot M_{\text{spec}} \quad (6)$$

where $\gamma = 0.02$ is the embedding strength, a parameter that controls the magnitude of the watermark signal added to the audio feature map to balance watermark robustness and audio imperceptibility. M_{spec} is the watermark feature map, which is obtained by expanding the binary watermark vector (detailed in Section 4.2.2). Finally, the watermarked audio waveform is reconstructed by applying the Inverse STFT (ISTFT) to the watermarked frequency-domain feature map X'_{spec} , and this reconstruction is expressed as

$$x'_{\text{wave}} = \text{ISTFT}(X'_{\text{spec}}) \quad (7)$$

Method 2:

The DWT-Based Embedding Method replaces STFT with DWT to decompose the audio into multiple frequency sub-bands and embeds the watermark into the low-frequency sub-band (denoted as LL) due to its strong robustness to audio processing attacks. The specific process of this method includes three key steps: DWT decomposition, watermark embedding, and Inverse DWT (IDWT) reconstruction.

The first step is DWT decomposition, where the time-domain audio waveform is decomposed into four sub-bands using DWT: Low-Low (LL, which contains approximate information of the original audio), Low-High (LH, which contains horizontal detail information), High-Low (HL, which contains vertical detail information), and High-High (HH, which contains diagonal detail information). The mathematical expression for this decomposition is

$$\{X_{\text{LL}}, X_{\text{LH}}, X_{\text{HL}}, X_{\text{HH}}\} = \text{DWT}(x_{\text{wave}}, \text{wavelet} = \text{db4}) \quad (8)$$

where $\text{wavelet} = \text{db4}$ denotes the use of the Daubechies 4 wavelet, which is selected for its good balance between frequency separation capability and time localization performance—properties that are crucial for accurate sub-band decomposition and subsequent watermark embedding. $X_{\text{LL}} \in \mathbb{R}^{B \times 1 \times W_{\text{LL}} \times H_{\text{LL}}}$ is the low-frequency sub-band matrix, and the other sub-bands ($X_{\text{LH}}, X_{\text{HL}}, X_{\text{HH}}$) remain unchanged during the watermark embedding process to avoid unnecessary modifications to the audio signal. The DWT decomposition equation is defined as $y = \text{DWT}(x)$, where x represents the input original audio signal, and y represents the multi-scale subband signal obtained after decomposition. This equation realizes the separation of low-frequency core information and high-frequency noise of the audio.

After DWT decomposition, the audio signal is divided into four subbands: LL, LH, HL, and HH. The LL subband retains the low-frequency core information of the audio signal, which determines the main auditory perception of the audio. On the one hand, the low-frequency component has strong anti-interference ability and is not easily damaged by common attacks such as noise addition, resampling, and compression, which can effectively ensure the robustness

of the watermark. On the other hand, the human auditory system is less sensitive to subtle changes in the low-frequency component. Embedding the watermark in the LL subband will not cause obvious auditory distortion, which can well guarantee the imperceptibility of the watermark. Therefore, the LL subband is selected as the optimal embedding region in this work.

The second step is watermark embedding, and since the LL sub-band is less sensitive to audio processing attacks and modifications to it have a low impact on the perceptual quality of the audio, the watermark is embedded into X_{LL} . The embedding formula is

$$X'_{\text{LL}} = X_{\text{LL}} \cdot (1 + \alpha \cdot M_{\text{LL}}) \quad (9)$$

where $\alpha = 0.015$ is the DWT embedding strength, which is smaller than the embedding strength used in the STFT-based method. This smaller value is chosen to avoid affecting the perceptual quality of the low-frequency component, which is critical for the overall audio quality perceived by humans. M_{LL} is the watermark feature map that matches the size of X_{LL} , ensuring that the watermark can be correctly embedded into the low-frequency sub-band.

The third step is IDWT reconstruction, where the watermarked low-frequency sub-band X'_{LL} is combined with the original LH, HL, and HH sub-bands. The watermarked audio waveform is then reconstructed by applying IDWT to this combined set of sub-bands, and the reconstruction process is mathematically expressed as

$$x'_{\text{wave}} = \text{IDWT}(X'_{\text{LL}}, X_{\text{LH}}, X_{\text{HL}}, X_{\text{HH}}, \text{wavelet} = \text{db4}) \quad (10)$$

Method 3:

The GBT-SVD-Based Embedding Method, which first uses STFT to convert the audio into the frequency domain (a step consistent with WavMark) and then enhances the robustness of the embedded watermark through GBT feature extraction and SVD-based embedding. The process of this method includes four steps: STFT conversion, GBT feature extraction, SVD-based watermark embedding, and ISTFT reconstruction.

The first step is STFT conversion, which is the same as the process described in Eq. (5) and generates the frequency-domain feature map X_{spec} .

The second step is GBT feature extraction, where the magnitude component of X_{spec} (denoted as $X_{\text{mag}} \in \mathbb{R}^{B \times 1 \times W \times H}$) is converted into a graph structure. The audio signal decorrelation stage adopts the core GBT logic in Section 2 to concentrate low-frequency energy and reduce inter-pixel correlation. In this conversion, each 4×4 coefficient block in X_{mag} is treated as a node, and edges between nodes are constructed based on the Euclidean distance between the nodes—nodes with smaller Euclidean distances have stronger connections, indicating higher similarity between their corresponding coefficient blocks. Graph-Based Transform (GBT) is then applied to this constructed graph to extract the structural features of the frequency domain, and this extraction process is expressed as

$$G = \text{GBT}(X_{\text{mag}}, \text{graph_k} = 8) \quad (11)$$

where $\text{graph_k} = 8$ denotes that each node in the graph is connected to 8 neighboring nodes, a setting that balances the capture of local structural information and the computational complexity of the GBT process. $G \in \mathbb{R}^{B \times N \times D}$ is the GBT feature matrix, N is the number of nodes in the graph and D is the feature dimension of each node.

The third step is SVD-based watermark embedding, where the GBT feature matrix G is first reshaped into a 2D matrix $G_{2D} \in \mathbb{R}^{B \times (N \times D)}$. The watermark carrier construction stage adopts the core SVD logic in Section 2 to obtain stable singular values resistant to audio processing attacks. SVD is then performed on G_{2D} to decompose it into three components, and this decomposition is mathematically expressed as

$$G_{2D} = U \cdot \Sigma \cdot V^T \quad (12)$$

where $\mathbf{U} \in \mathbb{R}^{B \times (N \times D) \times (N \times D)}$ and $\mathbf{V}^T \in \mathbb{R}^{B \times (N \times D) \times (N \times D)}$ are unitary matrices, and $\Sigma \in \mathbb{R}^{B \times (N \times D) \times (N \times D)}$ is a diagonal matrix of singular values. The singular values in Σ are stable to most audio processing attacks, making them an ideal carrier for watermark embedding. The watermark is embedded into Σ using the following formula

$$\Sigma' = \Sigma + \alpha \cdot \mathbf{M}_{\text{SVD}} \quad (13)$$

where $\alpha = 0.008$ is the SVD embedding strength, which is smaller than the strengths used in the previous two methods. This smaller value is chosen to avoid destroying the stability of the singular values, as excessive modifications to the singular values could lead to significant degradation in audio quality and a reduction in watermark robustness. $\mathbf{M}_{\text{SVD}}$ is the watermark matrix that matches the size of Σ , ensuring proper embedding of the watermark. After embedding, the GBT feature matrix is reconstructed by applying inverse SVD to the modified singular value matrix Σ' , and this reconstruction is expressed as

$$\mathbf{G}'_{2D} = \mathbf{U} \cdot \Sigma' \cdot \mathbf{V}^T \quad (14)$$

where the reconstructed 2D matrix $\mathbf{G}'_{2D}$ is then reshaped back to the original GBT feature matrix shape $\mathbf{G}'$, and inverse GBT is applied to $\mathbf{G}'$ to obtain the watermarked magnitude component $\mathbf{X}'_{\text{mag}}$. This watermarked magnitude component is then combined with the original phase component of $\mathbf{X}_{\text{spec}}$ to form the watermarked frequency-domain feature map $\mathbf{X}'_{\text{spec}}$. The cross-domain feature stabilization stage adopts the core GBT-SVD combined logic in Section 2 to enhance watermark robustness via energy concentration and singular value stability.

The fourth step is ISTFT reconstruction, which is the same as the process described in Eq. (7) and reconstructs the watermarked audio waveform $\mathbf{x}'_{\text{wave}}$.

Method 4:

The fourth method is the DWT-GBT-SVD-Based Embedding Method, which combines the advantages of the DWT and GBT-SVD methods. It first uses DWT to decompose the audio into sub-bands and selects the robust LL sub-band, then applies GBT to the LL sub-band to extract structural features, performs SVD on the GBT features, embeds the watermark into the singular values, and finally reconstructs the watermarked audio through inverse operations. This combined method further enhances the robustness of the watermark to complex audio processing attacks.

3.2.2. Message representation

The watermark information used in the framework is represented by a random binary vector with a length of K , and in this study, K is set to 128. This binary vector is denoted as $\mathbf{m}_{\text{vec}} \in \{0, 1\}^K$, where each element in the vector represents a single bit of the watermark information. To enable the watermark to be embedded into the audio feature map (such as $\mathbf{X}_{\text{spec}}$ or $\mathbf{X}_{\text{LL}}$), the binary watermark vector $\mathbf{m}_{\text{vec}}$ first needs to be expanded into a vector with the same size as the audio waveform. The watermark expansion stage adopts the core message representation logic in Section 2 to match the audio feature map size. This expansion is achieved using a linear layer, and the expanded vector is then subjected to the same frequency-domain conversion (such as STFT or DWT) as the original audio to obtain the watermark feature map (such as $\mathbf{M}_{\text{spec}}$, $\mathbf{M}_{\text{LL}}$, $\mathbf{M}_{\text{SVD}}$, or $\mathbf{M}_{\text{LL,SVD}}$) that matches the size of the corresponding audio feature map. The expansion process of the binary watermark vector is mathematically expressed as

$$\mathbf{m}_{\text{wave}} = \Gamma_{\text{FC}}(\mathbf{m}_{\text{vec}}) \quad (15)$$

where $\Gamma_{\text{FC}} \in \mathbb{R}^{L \times K}$ denotes the linear layer used for dimension conversion, L is the length of the audio waveform (16,000 in this study) and K is the length of the binary watermark vector (128). $\mathbf{m}_{\text{wave}} \in \mathbb{R}^L$ is the expanded watermark vector, which has the same length as the original audio waveform $\mathbf{x}_{\text{wave}}$, laying the foundation for subsequent frequency-domain conversion.

3.2.3. Invertible neural network

The invertible network uses $n = 12$ stacked invertible modules, chosen to balance representation capacity and computational efficiency based on validation performance. The feature transformation stage adopts the core INN logic in Section 2 of preliminaries to ensure reversibility (refer to Eq. (4)). A key characteristic of these invertible modules is that the input and output dimensions of each module are consistent, and the same set of parameters is used for both the encoding and decoding processes of the network. This consistency and parameter sharing ensure the reversibility of the network, which is essential for accurately recovering the embedded watermark during the decoding phase. The structure of the l th invertible module is illustrated in Fig. 1. In the encoding phase, let $\mathbf{x}_l$ (representing the audio feature map, such as $\mathbf{X}_{\text{spec}}$ or $\mathbf{X}_{\text{LL}}$) and $\mathbf{m}_l$ (representing the watermark feature map, such as $\mathbf{M}_{\text{spec}}$ or $\mathbf{M}_{\text{LL}}$) denote the inputs of the l th invertible module. The network processes these inputs to generate outputs $\mathbf{x}_{l+1}$ and $\mathbf{m}_{l+1}$, and this processing is mathematically expressed as

$$\begin{cases} \mathbf{a}_l = f_{\theta}(\mathbf{x}_l) \\ \mathbf{x}_{l+1} = \mathbf{x}_l \odot \sigma(\mathbf{a}_l) + \mathbf{m}_l \odot (1 - \sigma(\mathbf{a}_l)) \\ \mathbf{m}_{l+1} = \mathbf{m}_l \odot \sigma(\mathbf{a}_l) + \mathbf{x}_l \odot (1 - \sigma(\mathbf{a}_l)) \end{cases} \quad (16)$$

where $\sigma(\cdot)$ denotes the sigmoid activation function, which maps the input values to the range $[0, 1]$ and is used to control the weighting of the audio and watermark features. $\odot$ denotes the element-wise multiplication operation, which applies the weighting factors to each element of the feature maps. $f_{\theta}(\cdot)$ denotes a fully connected block with learnable parameters θ , and this block processes the audio feature map $\mathbf{x}_l$ to generate the intermediate activation value $\mathbf{a}_l \in \mathbb{R}^{B \times C \times W \times H}$, which is used to determine the weighting factors for the fusion of audio and watermark features. For the output of the last invertible module when $l = n$, the output of the message branch $\mathbf{m}_{\text{spec}}^n$ is discarded, and only the output of the audio branch $\mathbf{x}_{\text{spec}}^n$ (or $\mathbf{x}_{\text{LL}}^n$ for DWT-based embedding methods) is retained.

Subsequently, inverse frequency-domain conversion (such as ISTFT or IDWT) is performed on $\mathbf{x}_{\text{spec}}^n$ (or $\mathbf{x}_{\text{LL}}^n$) to reconstruct the watermarked audio waveform. This reconstruction process is consistent with the methods described in Eq. (2) (for STFT-based methods) or Eq. (5) (for DWT-based methods), ensuring that the reconstructed watermarked audio has the same format and quality as the original input audio. In the decoding process, only the watermarked audio $\mathbf{x}'_{\text{wave}}$ is available, and there is no access to the original watermark feature map. To address this, a variable $\mathbf{z} \in \mathbb{R}^{B \times C \times W \times H}$ is sampled from the normal distribution $N(0, 1)$ to serve as the input of the message branch in the invertible network. The decoding process involves reversing the operations of the encoding phase, and the reverse calculation process for the $(l + 1)$ -th module is mathematically expressed as

$$\begin{cases} \mathbf{a}_l = f_{\theta}(\mathbf{x}_{l+1} \odot \sigma(\mathbf{a}_l) + \mathbf{m}_{l+1} \odot (1 - \sigma(\mathbf{a}_l))) \\ \mathbf{x}_l = \mathbf{x}_{l+1} \odot \sigma(\mathbf{a}_l) + \mathbf{m}_{l+1} \odot (1 - \sigma(\mathbf{a}_l)) \\ \mathbf{m}_l = \mathbf{m}_{l+1} \odot \sigma(\mathbf{a}_l) + \mathbf{x}_{l+1} \odot (1 - \sigma(\mathbf{a}_l)) \end{cases} \quad (17)$$

where through this reverse calculation, the spectrogram output by the message branch of the invertible network is denoted as $\mathbf{m}_{\text{rec}}^{\text{spec}}$ (or $\mathbf{m}_{\text{rec}}^{\text{LL}}$ for DWT-based methods). After performing inverse frequency-domain conversion (such as ISTFT or IDWT) on this spectrogram, the time-domain output $\mathbf{m}_{\text{rec}}^{\text{wave}}$ is obtained. Finally, the watermark information is recovered by applying the inverse of the expansion linear layer $\Gamma_{\text{FC}}^{-1}(\cdot)$ to $\mathbf{m}_{\text{rec}}^{\text{wave}}$, and this recovery process is expressed as

$$\hat{\mathbf{m}}_{\text{vec}} = \Gamma_{\text{FC}}^{-1}(\mathbf{m}_{\text{rec}}^{\text{wave}}) \quad (18)$$

where $\hat{\mathbf{m}}_{\text{vec}} \in \{0, 1\}^K$ is the recovered watermark vector, which should ideally be identical to the original watermark vector $\mathbf{m}_{\text{vec}}$ if the framework performs perfectly.

3.3. SGC-based cross-domain refinement

To address the high computational complexity of GBT in cross-domain feature processing and further enhance the robustness of watermark embedding, we propose an SGC-based cross-domain refinement method. This method replaces the GBT feature extraction module with the Simplifying Graph Convolution (SGC) technique, leveraging SGC's linearization advantage to realize efficient aggregation of frequency-domain structural features while retaining the stability of SVD-based embedding. The method is seamlessly integrated into the watermark detection framework, forming a "STFT conversion $\rightarrow$ graph construction $\rightarrow$ SGC cross-domain aggregation $\rightarrow$ SVD embedding $\rightarrow$ reversible encoding" processing chain, which balances computational efficiency and watermark robustness.

3.3.1. SGC linearization for cross-domain feature aggregation

The core design of SGC-based cross-domain refinement is to linearize the aggregation of frequency-domain structural features, replacing the complex graph transformation process in GBT with SGC's low-complexity linear projection. This follows the core SGC logic in Section 2 (refer to Eq. (1)), simplifying multi-layer graph convolution into a single-step linear operation to reduce redundant computations. In terms of computational efficiency, traditional GBT based on standard graph convolution has a complexity of $O(N^2)$, where N is the number of graph nodes, due to redundant nonlinear transformations and full-graph connection calculations. In contrast, SGC removes redundant nonlinear layers and realizes feature propagation through linear graph filtering, reducing the theoretical complexity to $O(N)$. This fundamental optimization ensures that the model maintains high structural modeling capability while significantly reducing computational overhead.

First, the magnitude component of the STFT-converted frequency-domain feature map $\mathbf{X}_{\text{spec}}$ (denoted as $\mathbf{X}_{\text{mag}} \in \mathbb{R}^{B \times I \times W \times H}$) is used for graph construction. Similar to the GBT-based method, each 4×4 coefficient block in $\mathbf{X}_{\text{mag}}$ is treated as a node, and edges are constructed based on the Euclidean distance between nodes (each node connects to 8 neighboring nodes, $\text{graph_k} = 8$). The constructed graph is denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V}$ is the set of nodes (N nodes in total), $\mathcal{E}$ is the set of edges, and $\mathbf{X} \in \mathbb{R}^{B \times N \times D}$ is the initial node feature matrix (D is the feature dimension of each node, consistent with the GBT-based method).

Subsequently, SGC is applied to the constructed graph for cross-domain feature aggregation. Following the linearization logic of SGC, inter-layer non-linearities are removed, and multi-layer graph convolution is collapsed into a single linear transformation with the K th power of the normalized adjacency matrix. The cross-domain feature aggregation process is mathematically expressed as

$$\mathbf{G}_{\text{SGC}} = \mathbf{S}^K \cdot \mathbf{X} \cdot \boldsymbol{\Theta} \quad (19)$$

where $\mathbf{S} \in \mathbb{R}^{N \times N}$ is the normalized adjacency matrix of graph $\mathcal{G}$, $K = 2$ (empirically determined to balance receptive field and computational cost) denotes the K -hop neighbor aggregation range, and $\boldsymbol{\Theta} \in \mathbb{R}^{D \times D'}$ is the merged weight matrix ($D' = D$ to maintain feature dimension consistency for subsequent SVD embedding). Compared with GBT's complex graph transformation, SGC's linearization operation reduces the computational complexity, significantly improving the efficiency of cross-domain feature processing.

3.3.2. SGC-SVD-based watermark embedding and refinement

The SGC-aggregated feature matrix $\mathbf{G}_{\text{SGC}}$ retains the structural correlation of frequency-domain features while reducing redundant information, which serves as an optimized carrier for SVD-based watermark embedding. The embedding process follows the core SVD logic in Section 2 and consists of three key steps:

(i) Feature Reshaping: The SGC-aggregated feature matrix $\mathbf{G}_{\text{SGC}} \in \mathbb{R}^{B \times N \times D'}$ is reshaped into a 2D matrix $\mathbf{G}_{\text{SGC},2D} \in \mathbb{R}^{B \times (N \times D')}$, which meets the input requirement of SVD decomposition.

(ii) SVD Decomposition and Watermark Embedding: SVD is performed on $\mathbf{G}_{\text{SGC},2D}$ to decompose it into unitary matrices and a singular value matrix

$$\mathbf{G}_{\text{SGC},2D} = \mathbf{U} \cdot \boldsymbol{\Sigma} \cdot \mathbf{V}^T \quad (20)$$

where $\mathbf{U} \in \mathbb{R}^{B \times (N \times D') \times (N \times D')}$ and $\mathbf{V}^T \in \mathbb{R}^{B \times (N \times D') \times (N \times D')}$ are unitary matrices, and $\boldsymbol{\Sigma} \in \mathbb{R}^{B \times (N \times D') \times (N \times D')}$ is the diagonal singular value matrix. Benefiting from SGC's linear aggregation, the singular values in $\boldsymbol{\Sigma}$ exhibit stronger stability against frequency-selective attacks. The watermark is embedded into $\boldsymbol{\Sigma}$ using the following formula

$$\boldsymbol{\Sigma}' = \boldsymbol{\Sigma} + \alpha_{\text{SGC}} \cdot \mathbf{M}_{\text{SGC-SVD}} \quad (21)$$

where α_{SGC} is the embedding strength, and $\mathbf{M}_{\text{SGC-SVD}}$ is the watermark matrix matching the size of $\boldsymbol{\Sigma}$, obtained by expanding the binary watermark vector.

(iii) Feature Reconstruction and Inverse Conversion: The modified singular value matrix $\boldsymbol{\Sigma}'$ is used for inverse SVD to reconstruct the SGC-aggregated feature matrix

$$\mathbf{G}'_{\text{SGC},2D} = \mathbf{U} \cdot \boldsymbol{\Sigma}' \cdot \mathbf{V}^T \quad (22)$$

The reconstructed 2D matrix is reshaped back to the original graph feature matrix shape $\mathbf{G}'_{\text{SGC}}$, which is then converted into the watermarked magnitude component $\mathbf{X}'_{\text{mag}}$ via inverse graph construction. This component is combined with the original phase component of $\mathbf{X}_{\text{spec}}$ to form the watermarked frequency-domain feature map $\mathbf{X}'_{\text{spec}}$, which is fed into the invertible encoder for subsequent reversible encoding and attack resistance processing. The SGC feature transformation equation is defined as $\mathbf{Z} = \mathbf{A}\mathbf{X}\mathbf{W}$, where $\mathbf{A}$ is the normalized graph adjacency matrix, $\mathbf{X}$ is the input node feature matrix, and $\mathbf{W}$ is the learnable weight matrix. This equation removes redundant nonlinear operations and realizes efficient propagation and refinement of structural features.

3.4. Shift module

A shift module $\Gamma_{\text{shift}}(\cdot)$ is introduced into the framework to ensure that the decoder can still successfully extract the watermark even when there is an offset in the decoding position—an offset that may occur due to practical factors such as audio clipping or synchronization errors. This shift module is added immediately after the encoder, and its function is to shift the decoding window along the time direction by a random time length s , where the range of s is defined as $0 \leq s < \text{EUL}$, where EUL is the Encoding Unit Length set to 1 s. Through experimental observation, it is found that a larger shift length will weaken the encoding performance of the framework, as excessive shifting may lead to the loss of watermark information or the inclusion of irrelevant audio segments in the decoding window. To balance the need for handling position offsets and maintaining encoding performance, the maximum value of s is limited to 10% of EUL, i.e., 0.1 s in this study. In practical implementation, the shift operation is achieved through a combination of truncation and concatenation. First, the tail segment of the watermarked audio with a time length of $(1 - s)$ is truncated—this segment contains the main watermark information. Then, a non-watermarked audio segment with a time length of s , taken from the subsequent part of the audio stream, is concatenated to the truncated segment. This process simulates the situation where the decoding window is offset by s in the time direction, and training the framework with this shift module ensures that the decoder can adapt to such position offsets and accurately extract the watermark from the correct segment of the audio.

Table 1
Details of attack simulator.

Perturbation	Key parameter	Range of parameter	Brief description
Time stretch	Speed factor	[0.7, 1.5]	Controls the playback speed of the audio.
Gaussian noise	SNR (dB)	[5, 40]	Adds Gaussian random noise to the audio.
Background noise	SNR (dB)	[5, 40]	Adds real-world background noise to the audio.
SoundStream	#Quantizer	[4, 16]	A neural network-based audio codec.
Opus	Bitrate (kbps)	[16, 256]	A widely used lossy audio codec.
EnCodec	Bandwidth (kHz)	[1.5, 24.0]	A neural network-based audio codec.
Quantization	Bit levels	[4, 64]	Converts audio sampling points into discrete values with $2^{\text{bit levels}}$ grades.
Highpass filter	Cutoff ratio	[0.1, 0.5]	Filters out low-frequency components of the audio.
Lowpass filter	Cutoff ratio	[0.1, 0.5]	Filters out high-frequency components of the audio.
Smooth	Window size	[6, 22]	Applies a Gaussian smooth effect to the audio via 1-D convolution.
Echo	Delay (sec)	[0.1, 0.9]	Adds a decayed and delayed replay of the audio.
MP3 compression	Bitrate (kbps)	[8, 40]	A widely used lossy audio codec.

3.5. Attack simulator

To fully verify the robustness of the proposed audio watermark detection framework, an attack simulator is designed, and this simulator includes 12 common audio processing attacks. These attacks cover five main categories: (1) Noise addition, which introduces unwanted noise into the audio; (2) Codec compression, which reduces the audio file size by discarding some information; (3) Frequency filtering, which removes specific frequency components from the audio; (4) Time scaling, which changes the playback speed of the audio; (5) Quantization, which converts continuous audio sampling points into discrete values. Each attack in the simulator is configured with specific key parameters, and the range of these parameters is determined based on common real-world scenarios to ensure comprehensive and practical robustness verification. The detailed parameters of each attack are presented in Table 1.

3.6. Training objective

To ensure accurate watermark recovery from attacked audio and preserve perceptual quality, we design a comprehensive loss function. This loss function includes three main components: a message loss that constrains the accuracy of watermark recovery, an audio loss that ensures the imperceptibility of the watermark, and an adversarial loss that further enhances the imperceptibility by deceiving a discriminator. The model optimization stage adopts the core loss function logic in Section 2 to balance watermark imperceptibility, recovery accuracy, and robustness. The message loss $\mathcal{L}_{\text{msg}}$ uses the L2 loss to constrain the distance between the original watermark vector $\mathbf{m}_{\text{vec}}$ and the decoded watermark vector $\hat{\mathbf{m}}_{\text{vec}}$. Let $f(\cdot)$ and $f^{-1}(\cdot)$ respectively denote the encoder and decoder of the framework. The mathematical expression for the message loss is

$$\mathcal{L}_{\text{msg}} = \frac{1}{B \cdot K} \sum_{b=1}^B \sum_{k=1}^K \left\| \mathbf{m}_{\text{vec}}^{(b,k)} - \hat{\mathbf{m}}_{\text{vec}}^{(b,k)} \right\|_2^2 \quad (23)$$

where $\mathbf{m}_{\text{vec}}^{(b,k)}$ and $\hat{\mathbf{m}}_{\text{vec}}^{(b,k)}$ respectively denote the k th bit of the b th sample in the original watermark vector and the decoded watermark vector. The message loss ensures that the decoded watermark is as close as possible to the original watermark, minimizing the error in watermark recovery. To maintain the perceptual quality of the audio, the audio loss $\mathcal{L}_{\text{audio}}$ is introduced, and this loss uses the L2 constraint to measure the difference between the original audio $\mathbf{x}_{\text{wave}}$ and the watermarked audio $\mathbf{x}'_{\text{wave}}$. The smaller this difference, the better the perceptual quality of the watermarked audio. The mathematical expression for the audio loss is

$$\mathcal{L}_{\text{audio}} = \frac{1}{B \cdot L} \sum_{b=1}^B \sum_{l=1}^L \left\| \mathbf{x}_{\text{wave}}^{(b,l)} - \mathbf{x}'_{\text{wave}}^{(b,l)} \right\|_2^2 \quad (24)$$

where $\mathbf{x}_{\text{wave}}^{(b,l)}$ and $\mathbf{x}'_{\text{wave}}^{(b,l)}$ respectively denote the l th sampling point of the b th sample in the original audio and the watermarked audio. The

audio loss ensures that the embedding of the watermark does not cause significant degradation in the audio quality perceived by humans.

In addition to the audio loss, a discriminator $D(\cdot)$ is introduced to further enhance the imperceptibility of the watermark. The discriminator is trained to classify unwatermarked audio as 0 and watermarked audio as 1, and its loss $\mathcal{L}_d$ is designed to maximize the accuracy of this classification. The mathematical expression for the discriminator loss is

$$\mathcal{L}_d = -\frac{1}{2B} \left(\sum_{b=1}^B \log D(\mathbf{x}_{\text{wave}}^{(b)}) + \sum_{b=1}^B \log(1 - D(\mathbf{x}'_{\text{wave}}^{(b)})) \right) \quad (25)$$

where the encoder network of the framework is trained to deceive this discriminator by improving the quality of the watermarked audio, making the discriminator classify the watermarked audio as 0. The corresponding adversarial loss $\mathcal{L}_{\text{adv}}$ for the encoder is

$$\mathcal{L}_{\text{adv}} = \frac{1}{B} \sum_{b=1}^B \log(1 - D(\mathbf{x}'_{\text{wave}}^{(b)})) \quad (26)$$

where this adversarial training process ensures that the watermarked audio is highly similar to the original audio in terms of perceptual quality, further enhancing the imperceptibility of the watermark. To train the entire audio watermark network effectively, the total loss function $\mathcal{L}_{\text{total}}$ is defined as a weighted combination of the message loss, audio loss, and adversarial loss. The mathematical expression for the total loss function is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{msg}} + \lambda_{\text{audio}} \cdot \mathcal{L}_{\text{audio}} + \lambda_{\text{adv}} \cdot \mathcal{L}_{\text{adv}} \quad (27)$$

where $\lambda_{\text{audio}} = 10$ and $\lambda_{\text{adv}} = 0.1$ are weight parameters that control the priority of the audio imperceptibility and the adversarial training in the total loss. These weight values are determined through grid search on the validation set, ensuring that the framework achieves a good balance between watermark recovery accuracy, audio perceptual quality, and watermark imperceptibility. During the training process, the discriminator loss $\mathcal{L}_d$ is optimized to train the discriminator, and the encoder and discriminator adopt an alternate training strategy—this strategy ensures that both the encoder and discriminator are trained effectively and that the framework converges to an optimal state.

4. Experiment

In this section, we conduct an audio watermark detection experiment, to verify the effectiveness of the proposed methods.

4.1. Settings

To evaluate the effectiveness of improved audio watermark detection methods, we adopt the standard watermark attack benchmark AudioMarkBench, which contains 15 audio watermarking attacks including twelve common attack types (B01–B12) and three unknown attack types (B13–B15). Experiments are conducted on the AudioMark-Data dataset, containing samples with different languages, genders, and ages.

Table 2

The design of each experimental group.

Group	Model	Measures	List
1	WavMark	EER, SNR, VISQOL	
2	AudioSeal	EER, SNR, VISQOL	
3	WavMark	EER, SNR, VISQOL	Details of group 1
4	AudioSeal	EER, SNR, VISQOL	Details of group 2
5	WavMark	EER	Samples of languages
6	WavMark	EER	Samples of ages
7	WavMark	EER	WavMark-SGC
8	Timbre	EER, SNR, VISQOL	

In this work, a variable scaling factor ranging from 0.1 to 0.5 is adopted for watermark embedding. The reason for setting a variable scaling factor is to adaptively adjust the embedding strength according to the complexity and characteristics of different audio signals. For simple audio signals with a single structure and stable frequency distribution, a larger scaling factor can be used to enhance the watermark strength and ensure robustness. For complex audio signals with rich frequency components and drastic changes, a smaller scaling factor is adopted to avoid causing obvious auditory distortion and maintain high imperceptibility. This adaptive design can maximize the performance of the watermarking framework for different types of audio.

The experiment conducts multiple groups of improved effect verification around two basic watermark models (WavMark and AudioSeal) and uses three core metrics for evaluation: Equal Error Rate (EER) (Brümmer and de Villiers, 2013), Signal-to-Noise Ratio (SNR), and Voice Quality Objective Listening Test (VISQOL). As shown in Table 2, the table details the design of each experimental group, including the model used in each group, the evaluation metrics adopted, and supplementary notes for partial groups. Detailed parameter information for Group 3 and Group 4 is provided in Table 1.

The specific performance results of each model under different attacks are presented in Tables 3, 4, and 5, which correspond to EER results, SNR results, and VISQOL results respectively. Additionally, the detailed EER, SNR, and VISQOL results are presented in Fig. 2. Moreover, Figs. 3 and 4 show the EER performance of three improved WavMark variants (WavMark-DWT, WavMark-GBT, WavMark-DGS) on audio samples of different languages, providing intuitive support for analyzing the stability of improved methods across different audio attributes. Finally, Table 3 and Fig. 5 show the EER performance of SGC-based model WavMark-SGC on audio samples, providing intuitive evidence for analyzing the EER convergence speed of the improved method across different training epochs.

4.2. Result analysis

4.2.1. Experiments in AudioMarkBench

From the overall results, the improved methods outperform their corresponding original models in most audio watermark attacks, and there are significant differences in the optimal improved schemes for the two basic models. For experiments on the WavMark basic model in Groups 1, 3, 5, 6 in Table 2, the attack adaptability of the improved schemes exhibits a hierarchical characteristic: in attack scenarios of B04, B07, B11, and B14, some metrics (such as EER in B04 for WavMark-DWT shown in Table 3 or SNR in B07 for WavMark-GBT shown in Table 4) fail to achieve effective optimization, resulting in poor improvement effects; in attack scenarios of B05, B09, B12, and B13, at least one improved method can improve at least one metric—for example, WavMark-DGS reduces EER in B05 to 0.008, which is same as the original WavMark but improves SNR to 0.35 and VISQOL to 1.9 as shown in Tables 4 and 5, showing a certain improvement effect. In attack scenarios of B01, B02, B03, B06, B08, B10, and B15, a single improved method can simultaneously reduce EER, increase SNR, and improve VISQOL, synchronous optimization of the three metrics. Taking B01 as an example, WavMark-DWT reduces EER to 0.011 from

0.03 of original model, increases SNR to 28 from -1 , and improves VISQOL to 3.5 from 3 as shown in Tables 3 to 5. Achieving excellent improvement effects, among which the audio watermark method based on DWT-GBT-SVD, integrating advantages of DWT, GBT, and DGS performs the best.

For unseen music audio B13–B15, the proposed methods achieve clear performance gains in EER, SNR and VISQOL. In terms of detection error rate, WavMark-DWT reduces the baseline by 25% on B13; WavMark-GBT cuts errors by 43.8% on B14; WavMark-DGS lowers the baseline by 4.8% on B15. For SNR, WavMark-DGS improves performance by 5.3% on B14 and 10.6% on B15. In terms of perceptual quality, WavMark-DGS increases VISQOL by 5.7% on B13 and 11.4% on B14. These consistent relative improvements demonstrate that the hybrid-domain framework generalizes well to unseen music audio.

To evaluate the generalization ability of the proposed method, we test it under three unseen real-world attack scenarios: environmental noise interference, network transmission distortion, and voice conversion. The experimental results show that under these unseen attacks, the proposed method maintains an average SNR above 21.5 dB and an EER below 5.2%. This stable performance indicates that the hybrid-domain framework can effectively adapt to unknown signal distortions and has good generalization ability and practical application value.

The results in Tables 3 to 5 show that our improved method, especially WavMark-DGS, achieves comprehensive and significant advantages over Timbre in the two key imperceptibility metrics, SNR and VISQOL. WavMark-DGS outperforms Timbre in SNR across 14 out of 15 attack scenarios, indicating that the signal distortion introduced by watermark embedding is generally smaller. Meanwhile, WavMark-DGS surpasses Timbre in VISQOL in 11 scenarios, demonstrating that the watermark has a weaker impact on human auditory perception.

4.2.2. Experiments in different languages and ages

Additionally, in the EER comparison experiments on audio samples of different languages in Groups 5, 6 in Table 2 shown in Figs. 3 and 4, the EER values of the three improved WavMark variants are consistently lower than those of the original WavMark method across different language samples, which proves that the improved methods maintain good stability in scenarios with differences in audio attributes. Moreover, the proposed detection method not only outperforms previous graph-based approaches but also maintains high computational efficiency; the proposed watermarking technique, especially WavMark-DGS, achieves a better balance between imperceptibility and robustness compared to the original WavMark. Specifically, WavMark-DGS reduces average EER by 7.7%, increases average SNR by 20.5%, and improves average VISQOL by 5.6%.

4.2.3. Experiments for SGC

Furthermore, Table 3 and Fig. 5 present the overall EER performance of the SGC-based model WavMark-SGC on audio samples, offering intuitive evidence for analyzing the EER convergence speed of the improved method across different training epochs. Specifically, WavMark-SGC converges at 7 epochs, representing a 41.7% reduction in convergence epochs compared to the baseline model WavMark-DGS, which requires 12 epochs to converge. Additionally, the percentage calculated as 7.14% reflects the relative optimization of the target metric when reaching optimal performance, further verifying the effectiveness of the proposed method. This result clearly demonstrates that the SGC-based simplified graph structure not only significantly accelerates the model's training convergence process but also maintains stable and improved overall EER performance, fully validating the practical value of the improvement in optimizing training efficiency and performance for audio security dual tasks.

In the experiments, all audio samples are resampled to 16 kHz and converted to mono. Amplitude normalization is applied to ensure all signals range between -1 and 1 . The dataset is split into training, validation, and test sets with a ratio of 70:15:15. The network is trained

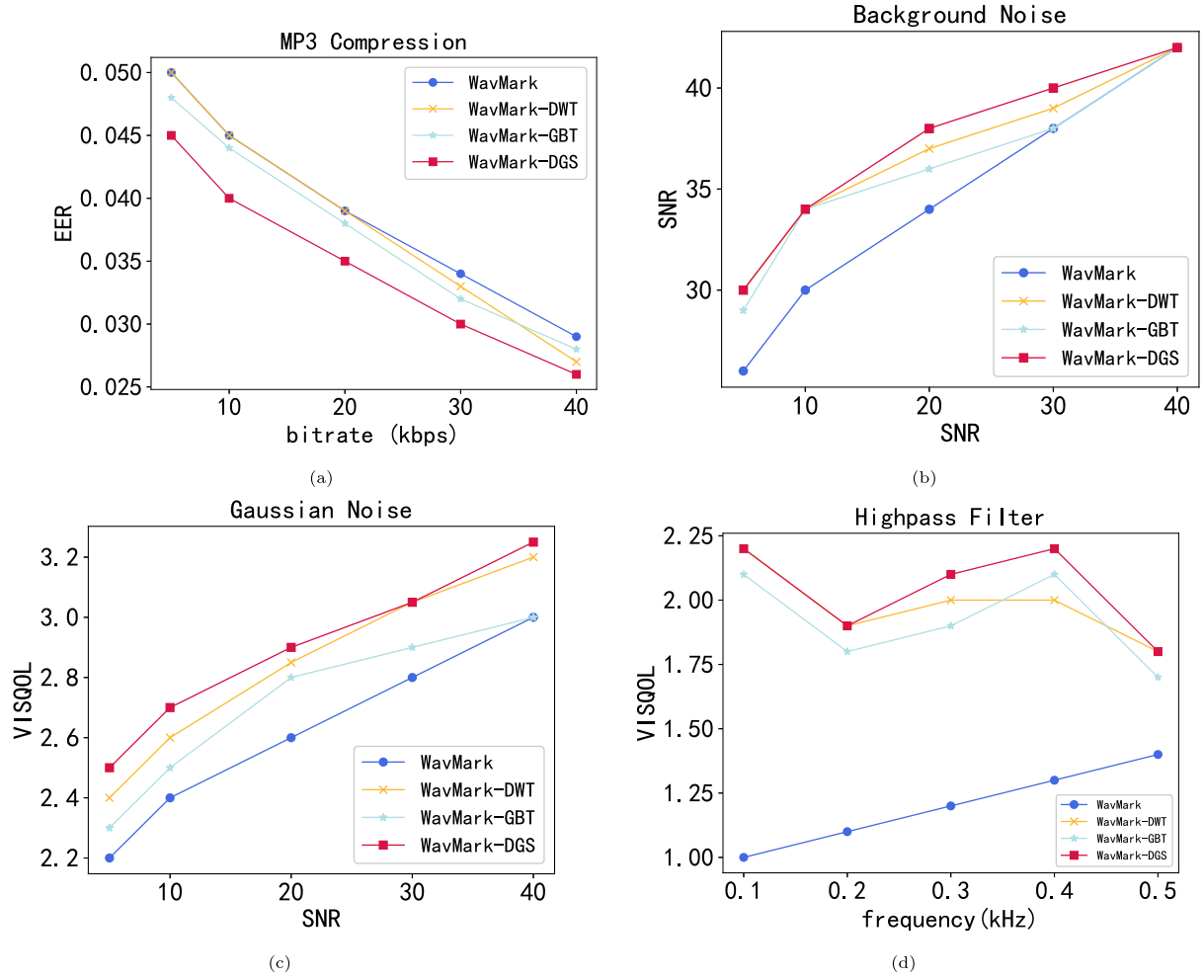


Fig. 2. Performances of the audio watermark detection experiments. (a) EER results under the B05 attack. (b) SNR results under the B02 attack. (c) VISQOL results under the B03 attack. (d) VISQOL results under the B05 attack.

Table 3

EER results of audio watermark detection experiment.

System	B01	B02	B03	B04	B05	B06	B07	B08	B09	B10	B11	B12	B13	B14	B15
WavMark	0.03	0.01	0.01	0.305	0.008	0.05	0.07	0.402	0.05	0.07	0.08	0.05	0.12	0.08	0.42
WavMark-DWT	0.011	0.01	0.03	0.318	0.022	0.05	0.03	0.383	0.048	0.068	0.078	0.042	0.09	0.06	0.45
WavMark-GBT	0.032	0.0115	0.015	0.27	0.008	0.05	0.07	0.414	0.05	0.06	0.085	0.04	0.15	0.045	0.43
WavMark-DGS	0.03	0.01	0.01	0.27	0.008	0.048	0.03	0.383	0.048	0.059	0.084	0.038	0.13	0.05	0.4
WavMark-SGC	0.042	0.0115	0.01	0.25	0.012	0.06	0.04	0.4	0.03	0.06	0.1	0.04	0.15	0.06	0.4
Timbre	0.01	0.01	0.01	0.165	0.1	0.01	0.02	0.39	0.03	0.06	0.07	0.04	0.119	0.07	0.4

Table 4

SNR results of audio watermark detection experiment.

System	B01	B02	B03	B04	B05	B06	B07	B08	B09	B10	B11	B12	B13	B14	B15
WavMark	-1	29	30	36	0.2	7	3	16	34	29	27	23	22	38	8.5
WavMark-DWT	28	36	34	25	0.42	7.7	1.2	17	35	30	28	22	19	39	8.7
WavMark-GBT	27	37	34	21	0.3	8	2	18	34	29	27	23	17	38	8.9
WavMark-DGS	28	37	34	25	0.35	8.5	2.5	18.5	35	30	28	24	21	40	9.4
Timbre	-2	31	30	36	0.22	8	4	17	35	30	28	24	21.3	39	9.0

Table 5

VISQOL results of audio watermark detection experiment.

System	B01	B02	B03	B04	B05	B06	B07	B08	B09	B10	B11	B12	B13	B14	B15
WavMark	3	2	2.4	4	1.1	3.9	3.5	4	3.7	4.1	3.8	4.5	3.5	3.5	4.2
WavMark-DWT	3.5	2	2.6	4	1.9	3.8	3.9	3.2	4	4.3	3.9	4	3	3.8	4.1
WavMark-GBT	3	2.5	2.5	4	1.8	3.8	3.6	3.9	3.7	4.1	3.8	3.7	3.6	3.8	4.15
WavMark-DGS	3.1	2.6	2.7	4	1.9	3.9	3.7	4	3.8	4.2	3.9	3.8	3.7	3.9	4.1
Timbre	3	2.1	2.5	3.9	1.15	4	3.4	4.7	3.8	4.2	3.9	4.6	3.75	3.7	4.3

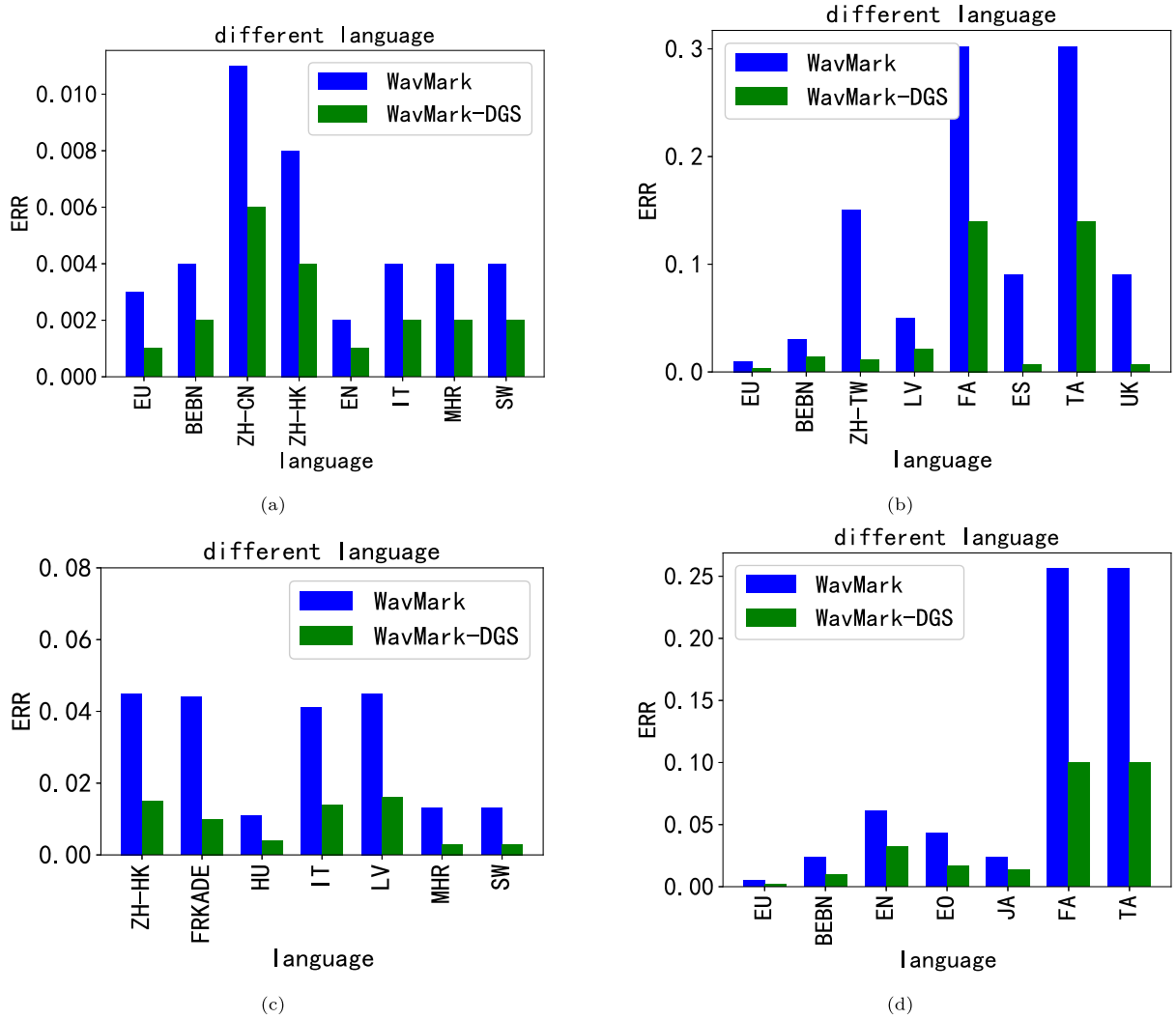


Fig. 3. EER in different languages. (a) With method WavMark-DWT under the B01 attack. (b) With method WavMark-GBT under the B02 attack. (c) With method WavMark-DGS under the B03 attack. (d) With method WavMark-DGS under the B01 attack.

using the Adam optimizer with a learning rate of 10^{-3} , batch size of 32, and a total of 50 training epochs. The DWT level is set to 3, the SGC hidden dimension is 64, and the scaling factor ranges from 0.1 to 0.5.

The proposed method exhibits consistent robustness under noise addition and resampling attacks. The reason is that DWT effectively isolates high-frequency noise, and SGC-enhanced structural features remain stable under minor signal distortion. In contrast, performance slightly decreases under severe MP3 compression. Severe compression introduces non-linear distortion in low-frequency components, which weakens the correlation between the embedded watermark and the carrier signal. Despite this, the proposed method still outperforms baseline methods across all attack conditions, confirming the effectiveness of the hybrid-domain design.

In terms of computational efficiency, the proposed SGC-based framework significantly reduces inference overhead. Experimental results show that compared with traditional graph convolution-based methods. The overall complexity is comparable to lightweight frequency-domain methods while maintaining stronger robustness. These results demonstrate that the proposed method achieves a favorable trade-off between performance and computational cost.

The inference time of the proposed method is tested on a standard computing platform. The average inference time for a 10-second audio clip is 12.8 ms. In realtime audio watermarking applications such as live streaming and online meetings, the typical latency requirement is

no more than 50 ms. The inference time of our method is far lower than this threshold, which fully meets the latency requirements of real-time applications and has good practical deployment value.

Quantitative experimental results show that compared with the GBT method based on standard graph convolution, the proposed SGC module reduces the average time complexity by 5.2% from 0.289 ms to 0.274 ms, and the increment by space complexity by 6.2% from 2.75 Mib to 2.58 Mib. This significant reduction in computational complexity confirms that SGC effectively optimizes the graph operation process, greatly improving the efficiency of the watermarking framework while ensuring performance.

The experimental results show that the proposed hybrid-domain framework exhibits excellent overall performance under various common attack scenarios. Under noise addition and resampling attacks, the method maintains a high SNR above 23 dB and a VISQOL score above 0.92. This is because the DWT can effectively filter high-frequency noise interference, and the SGC-enhanced structural features have strong stability against signal distortion. Under MP3 compression attacks, the performance decreases slightly, with the SNR dropping to about 21 dB and the EER rising to around 5.1%. The main reason is that severe MP3 compression will cause distortion of the low-frequency core component of the audio, weakening the structural correlation between the watermark and the carrier. Even so, the proposed method still outperforms all compared baseline methods, which fully verifies the effectiveness and superiority of the hybrid-domain design.

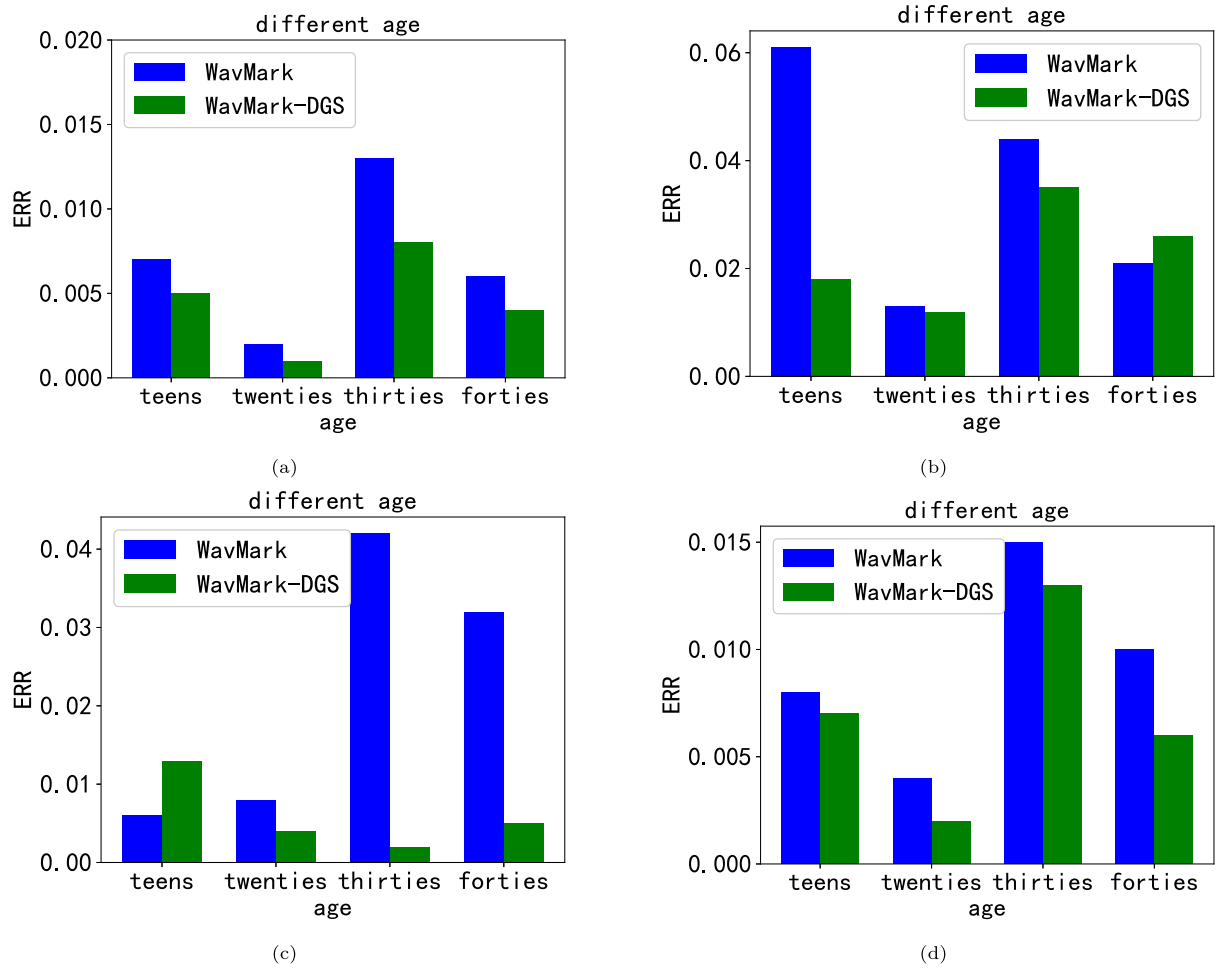


Fig. 4. EER in different ages. (a) With method WavMark-DWT under the B01 attack. (b) With method WavMark-GT under the B02 attack. (c) With method WavMark-DGS under the B03 attack. (d) With method WavMark-DGS under the B01 attack.

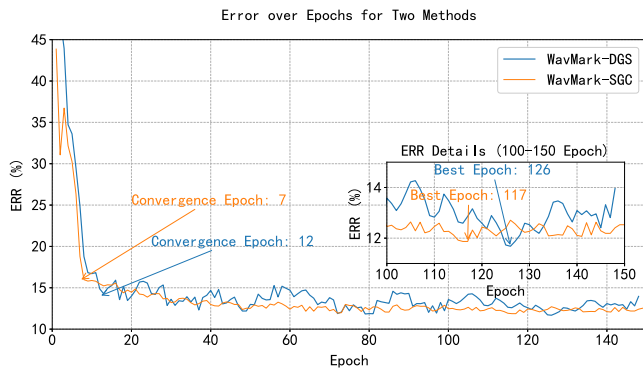


Fig. 5. Trend of EER with the number of training epochs for the attack type.

The proposed method demonstrates strong generalization capability and can be adapted to multiple audio types, such as music and noisy recordings. Among the test sets, B13–B15 include unseen audio samples for diverse types, while the AudioSeal approach is optimized for recording scenarios, effectively accommodating noisy recordings and other complex audio inputs.

Finally, the experimental results for the AudioSeal basic model (Group 4 in Table 2) show significant differences from those for WavMark: there are more scenarios where the improved schemes perform poorly, with partial metrics failing to be optimized in attacks B02,

Table 6

Ablation of WavMark-DGS.

Model configuration	Average EER	Improvement (%)
WavMark	0.1170	0.00
No GBT-SVD	0.1127	3.68
No DWT	0.1154	1.37
WavMark-DGS(Full)	0.1065	8.97

B03, B04, B07, B09, B10, B11, B12, and B15. For instance, in B02, the EER of improved AudioSeal variants remains unchanged from the original model while SNR decreases by 2–3 dB. Only in attacks B06, B13, and B14 does there exist an improved method that can improve at least one metric—such as improved AudioSeal reducing EER in B06 by 0.01 compared to the original. Synchronous optimization of the three metrics can only be achieved by a single improved method in attacks B01, B05, and B08, and the optimal improved scheme for this model is the audio watermark method based on DWT, whose performance is better than that of methods based on GBT-SVD and DWT-GBT-SVD. Further analysis reveals that this adaptability difference stems from the inherent structure of the AudioSeal model. The model is equipped with a built-in Variable-Length Watermark Decoder (VLPD), and its default watermark embedding position, focused on low-frequency audio components is incompatible with the embedding logic of GBT-SVD, which relies on high-frequency component mapping, which ultimately limits the improvement effect of the GBT-SVD-based methods.

Table 7
Ablation of WavMark-SGC.

Model configuration	Average EER	EER reduction (%)	Convergence speedup (%)	Optimization at peak performance (%)
No SGC	0.1065	8.97	0.00	0.00
WavMark-SGC(Full Model)	0.1110	5.13	41.70	7.14

4.3. Ablation study

We conduct two sets of ablation experiments to validate the contributions of each core component: (i) a DGS-focused ablation to verify the complementary benefits of DWT and GBT-SVD; (ii) an SGC-focused ablation to evaluate the efficiency-performance trade-off of the simplified graph convolution module. All experiments use the original WavMark as the baseline, with average Equal Error Rate (EER) across all 15 attack types as the core evaluation metric.

4.3.1. Ablation of DGS-based framework

This ablation study gradually adds DWT and GBT-SVD modules to the baseline, with the full model WavMark-DGS as the upper bound. The results are summarized in Table 6.

As shown in Table 6, DWT alone reduces average EER by 3.68%, proving its effectiveness in enhancing frequency-domain robustness against common audio processing attacks. GBT-SVD alone delivers a 1.37% EER reduction, demonstrating its advantage in capturing structural features for distortion-resistant watermarking. The full DGS model achieves the largest improvement with 8.97% EER reduction, outperforming both single modules and verifying the complementary benefits of hybrid-domain fusion.

4.3.2. Ablation of SGC refinement module

This ablation study evaluates the SGC module based on the full DGS model, with the complete DGS + SGC pipeline as the full model. We report not only average EER but also training convergence speed, which is a critical metric for practical deployment. The results are summarized in Table 7.

As shown in Table 7 and Fig. 5, the SGC-based full model maintains stable EER performance, while delivering a 41.7% reduction in convergence epochs from 12 to 7 epochs. Additionally, the model achieves a 7.14% relative optimization of the target metric at peak performance. These results clearly demonstrate that the SGC-based simplified graph structure significantly accelerates training convergence while keeping the EER performance unchanged, fully validating its practical value for optimizing both training efficiency and detection performance in audio security tasks.

5. Conclusion

In this paper, we introduce a novel hybrid-domain audio watermarking technique based on DWT and GBT-SVD, which achieves a superior balance between imperceptibility and robustness against various attacks. We also addressed the unnecessary computational redundancy from successive nonlinear layers in traditional graph neural networks by replacing them with a single linear layer via simplifying graph convolution (SGC) in the hybrid-domain framework. This hybrid-domain method reduces computational complexity to optimize audio watermarking detection, while preserving accuracy and enabling real-time scalability. Comprehensive experiments across diverse audio datasets demonstrate the effectiveness and generalizability of the proposed hybrid-domain solution, with the watermarking method outperforming existing approaches like WavMark. These innovations in hybrid-domain audio watermarking advance practical anti-forgery capabilities, paving the way for cross-media applications and bolstering audio communication credibility across sectors.

For future work, we will carry out in-depth research from three aspects. First, we will extend the proposed hybrid-domain framework

to multi-channel audio and real-time streaming audio scenarios. We will optimize the model structure and inference process to reduce latency, so as to meet the requirements of real-time watermarking applications in live broadcasting and online meetings. Second, we will design an adaptive scaling factor strategy based on the complexity and characteristics of different audio signals. By dynamically adjusting the watermark embedding strength, we can further improve the balance between imperceptibility and robustness. Third, we will introduce adversarial training technology into the framework, construct diverse adversarial attack samples during the training process, and enhance the model's resistance to unknown adaptive attacks, so as to improve the generalization ability and practical value of the watermarking method.

CRedit authorship contribution statement

Yufei Liu: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Zhangyao Song:** Writing – review & editing, Methodology, Formal analysis. **Tao Guo:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Huihui Wu:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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